

Álvaro Vilela Nova

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Madrid, Spain ♦ [Portfolio](#) ♦ [LinkedIn](#)

WORK EXPERIENCE

Multiverse Computing

December 2025 – Present

ML Engineer

Zaragoza, Spain

- End-to-end development and deployment of a legal-aware **agentic RAG** system using **LangGraph** and **Elasticsearch** ingesting tax documentation into hierarchical chunks with article, sub-article, and section metadata.
- Full design of the RAG evaluation pipeline using **LLM-as-a-judge** and **deterministic metrics**, prioritizing groundedness and correctness to minimize responses unsupported by the documentation.
- Deployment of client application on **AWS** using **EC2**, **Docker**, **FastAPI**, **PostgreSQL** and a web frontend, accessible via a whitelist-restricted VPN and supported by a CI/CD pipeline with tests and reproducible environments.

SOLUTE

August 2024 – December 2025

Data Scientist

Badajoz, Spain

- Development and deployment of energy forecasting models in **Python** with **PyTorch**, **scikit-learn** and **TensorFlow** for day-ahead auctions, intraday auctions, and **Almost Real-Time** electricity demand forecasting, with **MAE** improvements of up to 10–20% over market alternatives.
- Data collection, cleaning, and processing with **Pandas**, **NumPy**, external APIs, **BeautifulSoup** and pipelines orchestrated with **Airflow** combining geospatial meteorological data and time series.
- Creation of dashboards in **Power BI** to monitor accuracy KPIs, validate model-to-model improvements, and communicate project progress to technical and non-technical stakeholders.
- Development of a real-time **Computer Vision** system with **YOLO**, **OpenCV** and stereo vision to detect birds in wind farms and estimate distances exceeding 450 m.

EDUCATION

Universidad Complutense de Madrid

September 2024

Master's Degree, Big Data, Data Science & AI

Madrid, Spain

- **Master's Thesis Award** for an audiovisual query application with **Speech-to-Text**, **RAG** and **LLMs**.
- **PySpark** for data processing in **Azure Databricks**.
- Implementation of **SQL (MySQL)** and **NoSQL (MongoDB)**.

Universidad de Extremadura

June 2024

Bachelor's Degree, Mathematics

Badajoz, Spain

- **Bachelor's Thesis** – Descartes' Theorem and its Relation to Special Relativity.
- 10 months of education and cultural immersion with an **Erasmus+** scholarship in Poland.

CERTIFICATIONS AND LANGUAGES

- **Languages:** Native Spanish; **Fluent English (C2)**.
- **Skills:** Python; PyTorch; TensorFlow; scikit-learn; LangGraph; LlamaIndex; Elasticsearch; AWS; Airflow; SQL; PostgreSQL; MySQL; MongoDB; OpenCV; Docker; FastAPI; CI/CD; uv.